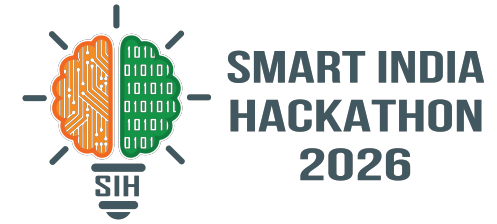


# SMART INDIA HACKATHON 2026

## TITLE PAGE



**Problem Statement ID – 26047**

**Problem Statement Title – Patient Case-Taking Software**

**Theme – Smart Automation**

**PS Category – Software**

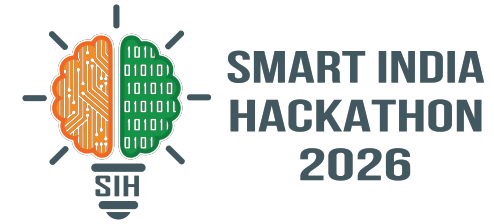
**Team ID – [Insert Team ID]**

**Team Name – [Insert Team Name]**



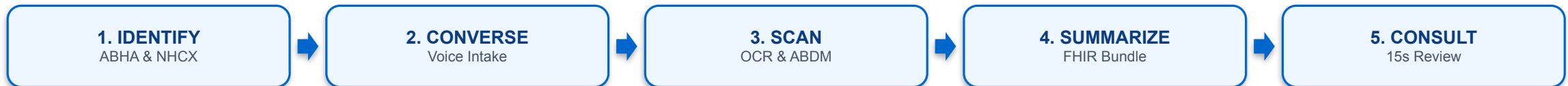
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# IDEA TITLE: MediKiosk



## MediKiosk — Sovereign Multimodal AI Clinical Intake & ABDM Digitization Platform

- **Autonomous Point-of-Entry Kiosk:** Deployed at OPD entry points to capture a patient's comprehensive medical history via voice + touch and digitize physical medical documents before seeing the doctor.
- **Solves 2.1-Min Consultation Crisis:** Overcomes India's severe OPD time collapse (BMJ Open, 2017) and the AYUSH Dashavidha Pariksha assessment gap by offloading structured history-taking to pre-consultation.
- **Dual Clinical Ontologies:** Integrates Western Allopathic SOCRATES framework (Site, Onset, Character, Radiation, Associations, Time, Exacerbating, Severity) and Ayurvedic Dashavidha Pariksha in a single engine.
- **Sub-150ms Emergency Red-Flag Interceptor:** Immediately flags acute cardiac, stroke, and respiratory distress symptoms for priority emergency triage rather than routine queueing.
- **100% Sovereign AI Foundation:** Powered by Sarvam AI (22 Indian languages STT, 11 languages TTS, 23 languages Document VLM)—all voice and document data processed sovereignly within India.
- **Doctor-in-the-Loop Safeguard:** Zero autonomous diagnosis—generates an editable, structured SOAP draft note allowing the physician to review complete history in 15 seconds and devote full consultation to clinical care.



## 1. Core Technology Stack Architecture:

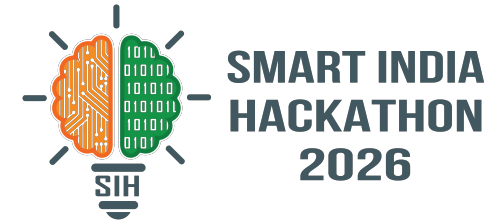
- **Frontend & Client:** Flutter 3.24 (patient kiosk UI with audio streaming & touch), Next.js 15 (physician consultation dashboard).
- **Backend & State:** FastAPI (Python 3.12 async gateway), LangGraph clinical state machine, Supabase/PostgreSQL 16 with Row-Level Security (RLS).
- **Sovereign AI Foundation:** Sarvam Saaras v3 (STT, 22 Indian languages), Sarvam Bulbul v3 (TTS, 11 languages), Sarvam Vision 3B VLM (Prescription/Lab OCR, 23 languages).
- **National Health Integration:** ABDM integration via Eka Care (NHA-certified HIP/HIU gateway for M1 ABHA, M2 HIP, M3 HIU), NHCX FHIR APIs for PM-JAY eligibility, and NRCeS-compliant HL7 FHIR R4 DocumentBundle output (bdl-11 invariant).

## 2. Methodology & 5-Step Clinical Implementation Pipeline:

- **Step 1 (Identify & Consent):** ABHA QR Scan & Share + DPDPA 2023 audio consent + NHCX PM-JAY benefit check.
- **Step 2 (Multimodal Converse):** Adaptive voice/touch interview via Sarvam Saaras + SOCRATES/Dashavidha tree + sub-150ms red-flag triage.
- **Step 3 (Scan & Fetch):** Prescription OCR via Sarvam 3B VLM + Eka Care M3 HIU longitudinal health record retrieval.
- **Step 4 (Summarize & Route):** Structured SOAP note generation + NRCeS FHIR R4 DocumentBundle linked to hospital HIS queue.
- **Step 5 (15-Sec Consult):** Physician reviews pre-compiled summary, edits/confirms in 15 seconds, and devotes consultation to care.

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# FEASIBILITY AND VIABILITY



## 1. Feasibility Analysis & Production Viability:

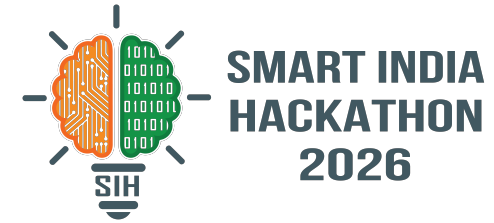
- **Production-Grade Stack:** Built on existing, commercially deployed sovereign APIs (Sarvam AI, Eka Care ABDM, NHCX)—requires systems integration, not unproven fundamental research.
- **Established ABDM Pathway:** Follows standardized NHA milestone pathways (M1 → M2 → M3) supported by official reference implementations (NHA ABDM-wrapper).
- **Modest Hardware Requirements:** Operates on standard off-the-shelf kiosk hardware (touchscreen, directional noise-cancelling mic, document camera), identical to airport/hospital check-in kiosks.

## 2. Potential Challenges & Risk Mitigation Strategies:

- ✓ **OPD Acoustic Noise ( $\geq 70\text{dB}$ ):** Mitigation: Dual-mode voice + high-contrast icon touch interface as seamless fallback when voice input is disrupted.
- ✓ **Low Digital Literacy:** Mitigation: Icon-driven, audio-guided conversational UI (Sarvam Bulbul) requiring zero prior training (validated for rural/elderly).
- ✓ **Handwriting OCR Variations:** Mitigation: Sarvam 3B VLM trained on Indian prescriptions + mandatory doctor-in-the-loop verification before saving.
- ✓ **Patient Trust & Consent:** Mitigation: Audio-guided regional explanation of DPDPA consent with cryptographic logging and ephemeral zero-trace memory.
- ✓ **NHA Gateway Latency:** Mitigation: Graceful offline/asynchronous fallback—kiosk completes intake locally even if ABDM/NHCX times out.

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# IMPACT AND BENEFITS



## 1. Target Audience Impact:

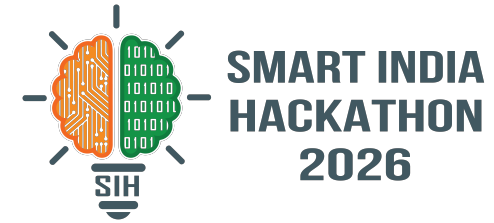
- **Patients:** Elderly, rural, low-literacy, and first-visit patients currently excluded by smartphone-dependent tele-triage apps gain effortless, audio-guided OPD access.
- **Physicians:** Doctor documentation time cut from ~3 minutes to a 15-second review, redirecting saved clinical time to physical examination and counselling.
- **AYUSH Institutions:** Enables full 10-parameter Dashavidha Pariksha intake that is currently impossible to conduct manually within high-volume OPD time constraints.

## 2. Multi-Dimensional Benefits:

- ★ **Social & Clinical Quality:** Eliminates diagnostic error from rushed history-taking; ensures sub-150ms emergency triage for critical conditions.
- ★ **Health-System Interoperability:** Generates standardized ABDM-linked FHIR R4 EHRs, permanently solving paper record fragmentation across multiple providers.
- ★ **Economic Scale:** Empowers high-throughput government OPDs (4,000–10,000 patients/day) to scale intake without hiring additional triage staff.
- ★ **Governance & Privacy:** 100% compliant with DPDPA 2023 consent protocols, NRCeS FHIR standards, and sovereign in-country data residency.
- ★ **Core Transformation:** Pre-consultation history time reduced by >90% (from 3 minutes to 15 seconds).

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# RESEARCH AND REFERENCES



1. **Primary Care Consultation Time Study:** Irving G, Neves AL, Dambha-Miller H, et al. "International variations in primary care physician consultation time: a systematic review of 67 countries." BMJ Open. 2017;7(10):e017902. DOI: 10.1136/bmjopen-2017-017902
2. **Diagnostic Value of History-Taking:** Hampton JR, Harrison MJ, Mitchell JR, Prichard JS, Seymour C. "Relative contributions of history-taking, physical examination, and laboratory investigation to diagnosis and management of medical outpatients." BMJ. 1975;2(5969):486-489. DOI: 10.1136/bmj.2.5969.486
3. **Classical Ayurvedic Diagnostic Framework:** Charaka Samhita, Vimana Sthana, Chapter 8 (Rogabhishagjitiya Vimanam, Verses 94–130) — Standard Dashavidha Pariksha clinical ontology.
4. **National Resource Centre for EHR Standards:** NRCeS India FHIR R4 Implementation Guide for Ayushman Bharat Digital Mission (ABDM). URL: [nrces.in/ndhm/fhir/r4/](https://nrces.in/ndhm/fhir/r4/)
5. **National Health Authority (NHA) Specifications:** ABDM Milestone 3 (M3\_V1) & NHCX FHIR Coverage Eligibility Guidelines. URL: [sandbox.abdm.gov.in](https://sandbox.abdm.gov.in)
6. **Data Privacy & Sovereign Regulation:** The Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023), Ministry of Law and Justice, Government of India. Gazette Notification Aug 11, 2023.
7. **Sovereign Indic AI Platform:** Sarvam AI Developer Documentation: Saaras v3 (STT), Bulbul v3 (TTS), Sarvam Vision 3B VLM. URL: [sarvam.ai](https://sarvam.ai)
8. **Certified National ABDM Gateway:** Eka Care ABDM Connect Developer Suite: NHA-Certified M1, M2, and M3 APIs. URL: [developer.eka.care](https://developer.eka.care)